SOFTWARE TEST PLAN

OrangeHRM Open Source Demo

Manual Testing Documentation Project

DOCUMENT CONTROL	DETAILS
Project type	Portfolio project
Application type	Human Resource Management System (HRMS)
Testing approach	Manual test planning and test case design
Document status	Planned coverage — test execution not performed
Prepared by	Arpan Sircar
Version	1.0

Application URL: https://opensource-demo.orangehrmlive.com/web/index.php/auth/login

DOCUMENT NOTE

This document defines proposed test coverage and sample test cases only. No test execution, actual defect discovery, pass/fail results, or application quality assessment is claimed.

1. Introduction

This document defines the planned manual testing scope for the OrangeHRM Open Source Demo application. It focuses on identifying potential test scenarios, designing sample test cases, and understanding foundational test design techniques.

Formal business requirements are not available. The planned scope is therefore based on visible application areas and expected behavior inferred from common user-interface conventions. Expected behavior should be confirmed against the application if testing is performed later.

2. Objectives

- Understand common workflows in an HRMS.
- Identify potential test scenarios from the available application interface.
- Prepare structured manual test cases with preconditions, steps, and expected results.
- Introduce fundamental test design techniques, including BVA and Equivalence Partitioning.
- Define a consistent approach for documenting defects if any are observed during future execution.

3. Project Scope

3.1 Features Considered for Test Planning

Area	Planned coverage
Login and authentication	Valid/invalid credential scenarios and required-field handling
Logout and session handling	Logout flow and possible expired-session behavior
Dashboard and navigation	Navigation to available modules
Employee information	Potential search, filtering, and form-validation scenarios where accessible
User management	Potential user-related workflows where accessible
Input validation	Required fields and documented input limits
UI and usability	Labels, validation messages, navigation consistency, and control readability

3.2 Out of Scope

- Features unavailable or inaccessible in the demo environment.
- Verification against formal business requirements.
- Database-level validation and API testing.
- Comprehensive security penetration testing.
- Load, stress, and production-environment testing.
- Actual defect verification, fix retesting, and resolution tracking.

4. Test Strategy

The project uses a risk-aware, manual test-planning approach. Scenarios are proposed based on common application behavior, visible functionality, and potential user impact. These activities describe planned coverage and are not evidence of completed testing.

Testing activity	Purpose	Illustrative example
Functional testing	Identify scenarios for checking available feature behavior.	Valid login should lead to the appropriate landing page.
Positive testing	Define cases using valid inputs and expected actions.	Submit the login form with valid credentials.
Negative testing	Define cases using invalid or missing inputs.	Submit the login form with an incorrect password.
Smoke testing	Select important scenarios for a basic availability check.	Consider login and primary navigation.
Regression testing	Identify scenarios that could be rerun after a change.	Rerun login and navigation cases after a change.
Exploratory testing	Plan areas to investigate beyond predefined steps.	Consider unusual input and navigation paths.
UI and usability	Consider consistency, labels, messages, and readability.	Review whether controls and messages are understandable.
Compatibility testing	Identify browser and screen-size combinations for later checks.	Plan comparison across supported browsers and viewport sizes.

5. Test Design Techniques

Technique	Purpose	Example
Boundary Value Analysis (BVA)	Identify values at and around a known input boundary.	If a documented maximum is 30 characters, consider 29, 30, and 31.
Equivalence Partitioning	Group inputs expected to behave similarly.	For a required field, consider empty and non-empty input classes.
Decision Table Testing	Consider combinations of conditions and expected outcomes.	Username valid/invalid combined with password valid/invalid.
State Transition Testing	Consider behavior as the application changes state.	Login page → authenticated session → logout.
Use Case Testing	Cover a typical user workflow.	Log in and navigate to an available module.
Error Guessing	Consider common failure conditions.	Submit a form with missing required values.
Checklist-Based Testing	Use a repeatable list of common checks.	Review labels, validation messages, and navigation.

Important: Specific boundary values must not be assumed until field limits are known. Examples explain the techniques and do not assert unverified OrangeHRM field rules.

6. Test Environment and Test Data

Item	Planned setup / note
Application	OrangeHRM Open Source Demo
URL	https://opensource-demo.orangehrmlive.com/web/index.php/auth/login
Browser	Google Chrome — intended environment
Operating system	Windows 11 — intended environment
Testing approach	Manual test planning; execution not performed
Test data	Illustrative valid and invalid inputs; no real employee data
Environment status	Browser compatibility and environment behavior have not been verified

Use only demo credentials and functionality that are publicly provided or permitted. Do not enter real employee information, confidential data, or personal information.

7. Sample Test Scenarios

Scenario ID	Feature	Planned scenario	Priority
TS-01	Login	Consider login with valid credentials	High
TS-02	Login	Consider login with an invalid password	High
TS-03	Login	Consider login with an empty username	High
TS-04	Login	Consider login with an empty password	High
TS-05	Authentication	Consider logout behavior	High
TS-06	Navigation	Consider navigation to available modules	Medium
TS-07	Employee management	Consider searching for an employee, if accessible	Medium
TS-08	Form validation	Consider submitting a form with missing required fields	Medium
TS-09	Input validation	Consider values near a documented field-length limit	Medium
TS-10	Session handling	Consider behavior after a session expires	Medium

These are proposed scenarios, not completed test results. Feature availability and expected behavior should be confirmed if execution is performed later.

8. Sample Test Cases

TC-01 — Login with valid credentials

Field	Details
Module	Login
Preconditions	Login page is accessible; valid demo credentials are available.
Test data	Valid demo username and password.
Test steps	1. Open the OrangeHRM login page. 2. Enter a valid username. 3. Enter the corresponding valid password. 4. Select Login.
Expected result	The application should authenticate the user and navigate to the appropriate landing page.
Execution result	Not recorded

TC-02 — Login with an invalid password

Field	Details
Module	Login
Preconditions	The login page is accessible.
Test data	Valid demo username and an incorrect password.
Test steps	1. Open the login page. 2. Enter a valid username. 3. Enter an incorrect password. 4. Select Login.
Expected result	The application should reject the login attempt and display an appropriate error without granting access.
Execution result	Not recorded

TC-03 — Login with an empty username

Field	Details
Module	Login
Preconditions	The login page is accessible.
Test data	Empty username and a password value.
Test steps	1. Open the login page. 2. Leave the username field empty. 3. Enter a password. 4. Select Login.
Expected result	The application should prevent successful authentication and handle

	the missing username appropriately.
Execution result	Not recorded

TC-04 — Boundary Value Analysis for a text field

Field	Details
Module	Input validation
Preconditions	The field's accepted length limit is known from reliable application information.
Test data	Values immediately below, at, and above the known boundary.
Test steps	1. Identify the documented field-length limit. 2. Prepare a value just below the limit. 3. Prepare a value at the limit. 4. Prepare a value just above the limit. 5. Define expected behavior for each value.
Expected result	Each value should be accepted or rejected according to the defined validation rules.
Execution result	Not recorded

9. Defect Reporting Approach

If a defect is observed during future test execution, the following fields can be used. No actual defects are claimed or recorded in this test planning project.

Field	Description
Defect ID	Unique identifier assigned when a defect is recorded
Title	Short, specific summary of the issue
Module	Affected application area
Preconditions	Conditions required to reproduce the issue
Steps to reproduce	Numbered steps that reliably reproduce the issue
Expected result	Intended behavior
Actual result	Observed behavior
Severity	Impact of the issue on functionality
Priority	Urgency with which the issue should be addressed
Evidence	Screenshot or other supporting evidence, if available
Status	Status according to the chosen workflow

10. Entry and Exit Criteria

10.1 Entry Criteria

- The demo application URL has been identified.
- The purpose and scope of the portfolio project are defined.
- Planned test areas and techniques are documented.
- A consistent sample test case format is established.

10.2 Completion Criteria for This Document

- The project objective and scope are documented.
- Planned testing activities and test design techniques are described.
- Representative sample scenarios and test cases are prepared.
- The defect reporting format is defined.
- The document clearly identifies test execution as Not Run.

These criteria apply only to completion of the documentation. They do not indicate that the application has passed testing or is ready for release.

11. Risks and Limitations

- Formal business requirements are not available.
- Some expected behaviors are inferred from common application conventions and remain unverified.
- Some features may be restricted by demo account permissions or may not be accessible.
- The demo environment and available functionality may change over time.
- Functionality, compatibility, performance, and security have not been verified through execution.
- The document does not establish that the application is defect-free.

12. Project Deliverables

Deliverable	Status / rationale
Manual test plan	Included in this document
Sample test scenarios	Included in this document
Sample test cases	Included in this document
Test design technique examples	Included in this document
Defect report template	Included as a field list

A Requirements Traceability Matrix is not included because formal requirements are unavailable. A Test Summary Report, execution metrics, and actual defect log are not included because test execution has not been performed.

13. Conclusion

This portfolio project demonstrates foundational knowledge of manual test planning and structured test documentation for a publicly available HRMS demo application. It covers scope definition, sample test scenarios,

representative test cases, basic test design techniques, and a defect reporting approach. All test cases remain unexecuted; therefore, no conclusions are made about the application's actual quality or behavior.